Portfolio Theory Notes

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0 Chapter map

5.1	**Risk**	measure risk with Var(K) or σ_K
5.2	**Two securities**	weights, μ_V, σ_V^2, the role of ρ_{12}, min-variance portfolio
5.3	**n securities**	matrix form ($\boldsymbol{m}, \boldsymbol{C}, \boldsymbol{w}, \boldsymbol{u}$), MVP, min-variance line, efficient frontier
5.4	**CAPM**	Capital Market Line, market portfolio, beta, Security Market Line

1 Notation cheat-sheet

Symbol	Meaning
K_i	(random) return on security i over one period
K_V	return on portfolio V
k	logarithmic return, $k = \ln(1 + K)$
$\mu_i = E(K_i)$, $\mu_V = E(K_V)$	expected returns
$\sigma_i = \sqrt{\text{Var}(K_i)}$, σ_V	standard deviations (risk)
$c_{ij} = \text{Cov}(K_i, K_j)$	covariance; $c_{ii} = \sigma_i^2$
$\rho_{12} = \dfrac{\text{Cov}(K_1, K_2)}{\sigma_1 \sigma_2}$	correlation coefficient, $-1 \le \rho_{12} \le 1$
x_i	**number of shares** of security i held
$S_i(0)$, $S_i(1)$	price of security i at time 0 and 1
$V(0)$, $V(1)$	portfolio value at time 0 and 1
w_i	**weight** = share of initial wealth in security i
s	parameter for two-security portfolios: $s = w_2$, $w_1 = 1 - s$
$\boldsymbol{w} = [w_1 \ \cdots \ w_n]$	row vector of weights
$\boldsymbol{m} = [\mu_1 \ \cdots \ \mu_n]$	row vector of expected returns
$\boldsymbol{u} = [1 \ \cdots \ 1]$	row vector of ones
$\boldsymbol{C} = [c_{ij}]$	$n \times n$ covariance matrix (symmetric, positive definite, assumed invertible)
r_F	risk-free return
M, μ_M, σ_M, K_M	market portfolio and its expected return / risk / return
β_V, α_V	beta (slope) and alpha (intercept) of the characteristic line
ε_V	residual random variable

In this book **vectors are row vectors**. So $\boldsymbol{m}\boldsymbol{w}^T$ is a scalar (dot product), and $\boldsymbol{w}\boldsymbol{C}\boldsymbol{w}^T$ is a scalar (quadratic form).

2 Section 5.1: Risk

2.1 What should "risk" capture?

Two things:

1. **How far** the possible returns are from some reference value, and
2. **How likely** each scenario is.

The **spread** alone (max − min) is a poor measure because it ignores probabilities:

- 22% w.p. 0.99 and 2% w.p. 0.01 → spread is 20%, but risk is tiny.
- 22% and 2% with probability 0.5 each → same spread, much riskier.

So we use the expectation $E(K)$ as the reference value and measure dispersion around it.

Risk measures

$$\mathrm{Var}(K) = E\Big[(K - E(K))^2\Big] = E(K^2) - E(K)^2, \qquad \sigma_K = \sqrt{\mathrm{Var}(K)}$$

σ_K is in the **same units** as returns (%), which makes it easier to interpret. Variance is in squared units.

Scaling rule

$$\mathrm{Var}(aK) = a^2 \,\mathrm{Var}(K), \qquad \sigma_{aK} = |a|\,\sigma_K$$

In practice. This is why leverage hurts so much. A 2× leveraged equity fund has 2× the volatility and 4× the variance of the underlying index.

2.2 Remark 5.1: why ordinary returns, not log returns?

- Over **many time steps**, log returns add up: $k(0,n) = k(1) + \cdots + k(n)$. If the steps are independent:
$$\mathrm{Var}(k(0,n)) = \mathrm{Var}(k(1)) + \cdots + \mathrm{Var}(k(n))$$
- In this chapter we hold **several securities over one step**. Ordinary returns combine **linearly** across securities (see 5.2), which is what we need. So the chapter uses K, not k.

3 Section 5.2: Two securities

3.1 Diversification in one example

Two stocks, two equally likely scenarios:

Scenario	Prob	K_1	K_2
ω_1	0.5	10%	−5%
ω_2	0.5	−5%	10%

Split 50/50 → you earn **5% in both scenarios**. The risk is **zero**, even though each stock on its own is risky. This works because the returns are **perfectly negatively correlated**.

In practice. An umbrella maker and an ice-cream maker. Each one's revenue depends on the weather, but owning both smooths out the weather risk. (Equity vs. gold in a crisis is a loose, imperfect version of the same idea.)

3.2 Weights

Weights (book eq. for w_1, w_2 and (5.1))

$$w_1 = \frac{x_1 S_1(0)}{V(0)}, \qquad w_2 = \frac{x_2 S_2(0)}{V(0)}, \qquad w_1 + w_2 = 1 \tag{5.1}$$

- w_k is the **fraction of the initial value** invested in security k.
- **Short selling** allowed → one weight can be **negative**, and the other is then > 1.
- Converting weights to share numbers: $x_i = w_i \dfrac{V(0)}{S_i(0)}$.

Weights drift. Weights are defined using **time-0 prices**. As prices move, the value split changes even if no shares are traded (Example 5.3: a 60/40 portfolio becomes 64.22/35.78).

3.3 Portfolio return

Proposition 5.1: portfolio return is the weighted average of the component returns

$$K_V = w_1 K_1 + w_2 K_2 \tag{5.2}$$

Proof idea: $V(1) = x_1 S_1(0)(1 + K_1) + x_2 S_2(0)(1 + K_2) = V(0)\Big(w_1(1 + K_1) + w_2(1 + K_2)\Big)$. Then divide by $V(0)$ and subtract 1.

Remark 5.2: the log-return version is *not* linear

$$e^{k_V} = w_1 e^{k_1} + w_2 e^{k_2} \tag{5.3}$$

3.4 Expected return and variance of a two-asset portfolio

Expected return (linearity of E)

$$E(K_V) = w_1 E(K_1) + w_2 E(K_2) \tag{5.4}$$

Theorem 5.2: portfolio variance

$$\mathrm{Var}(K_V) = w_1^2 \mathrm{Var}(K_1) + w_2^2 \mathrm{Var}(K_2) + 2 w_1 w_2 \,\mathrm{Cov}(K_1, K_2) \tag{5.5}$$

The same two formulas in compact notation:

The two workhorse formulas

$$\mu_V = w_1 \mu_1 + w_2 \mu_2 \tag{5.7}$$

$$\sigma_V^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \, \rho_{12} \, \sigma_1 \sigma_2 \tag{5.8}$$

- μ_V is **linear** in the weights. σ_V^2 is **quadratic** in them.
- The cross term $2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2$ is where diversification comes from. When $\rho_{12} < 1$, the portfolio's σ_V is **less** than the weighted average $w_1 \sigma_1 + w_2 \sigma_2$ (for long-only weights).

Illustration (Example 5.6 data)

Scenario	**Prob**	K_1	K_2
recession	0.4	−10%	20%
stagnation	0.2	0%	20%
boom	0.4	20%	10%

$\sigma_1^2 \approx 0.0184$, $\sigma_2^2 \approx 0.0024$, $\rho_{12} \approx -0.96309$.

Weights (w_1, w_2)	σ_V^2	**Compared with components**
(40%, 60%)	0.000736	**smaller than both**: diversification
(80%, 20%)	0.009824	between the two
(−50%, 150%)	0.0196	**larger than both**: shorting adds risk

Proposition 5.3: with no short selling, the portfolio is never riskier than its riskiest component

$$\sigma_V^2 \leq \max\{\sigma_1^2, \sigma_2^2\} \quad (\text{if } w_1, w_2 \geq 0)$$

Proof idea: $\rho_{12} \leq 1 \Rightarrow \sigma_V^2 \leq (w_1\sigma_1 + w_2\sigma_2)^2 \leq \max(\sigma_1, \sigma_2)^2$.

3.5 The extreme cases $\rho_{12} = \pm 1$: risk can be eliminated

Proposition 5.4: zero-risk portfolios

If $\rho_{12} = 1$ (and $\sigma_1 \neq \sigma_2$): $\sigma_V = 0$ for

$$w_1 = -\frac{\sigma_2}{\sigma_1 - \sigma_2}, \qquad w_2 = \frac{\sigma_1}{\sigma_1 - \sigma_2} \tag{5.9}$$

This **requires short selling** (one weight is negative).

If $\rho_{12} = -1$: $\sigma_V = 0$ for

$$w_1 = \frac{\sigma_2}{\sigma_1 + \sigma_2}, \qquad w_2 = \frac{\sigma_1}{\sigma_1 + \sigma_2} \tag{5.10}$$

No short selling needed (both weights are positive).

Why: with $\rho = \pm 1$, (5.8) becomes a perfect square, $\sigma_V^2 = (w_1\sigma_1 \pm w_2\sigma_2)^2$, so $\sigma_V = |w_1\sigma_1 \pm w_2\sigma_2|$, which you can set to zero.

In practice. $\rho = +1$ with shorting is the logic behind a **hedge**. Example: long a stock and short the right amount of its futures. Both move together, so the shorted position cancels the risk.

Parametric form ($s = w_2$, $w_1 = 1 - s$):

$$\rho_{12} = -1: \quad \sigma_V = |(1-s)\sigma_1 - s\sigma_2|, \qquad \mu_V = (1-s)\mu_1 + s\mu_2$$
$$\rho_{12} = +1: \quad \sigma_V = |(1-s)\sigma_1 + s\sigma_2|, \qquad \mu_V = (1-s)\mu_1 + s\mu_2$$

On the (σ, μ) plane: for $\rho = -1$ the line **bends to touch the μ-axis**. For $\rho = +1$ it is a **straight segment** between the two securities. $0 \leq s \leq 1$ means no short selling.

3.6 General ρ_{12}: the portfolio line and the minimum-variance portfolio

Two-security portfolio line in terms of $s = w_2$

$$\mu_V = (1-s)\mu_1 + s\mu_2 \tag{5.11}$$
$$\sigma_V^2 = (1-s)^2\sigma_1^2 + s^2\sigma_2^2 + 2s(1-s)\rho_{12}\sigma_1\sigma_2 \tag{5.12}$$

σ_V^2 is a parabola in s, opening upward (its s^2-coefficient is $\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2 > (\sigma_1 - \sigma_2)^2 \geq 0$).

Theorem 5.5: minimum-variance weight (for $-1 < \rho_{12} < 1$)

$$s_0 = \frac{\sigma_1^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2} \tag{5.13}$$

With no short selling, clip s_0 to $[0, 1]$:

$$s_{\min} = \begin{cases} 0 & s_0 < 0 \\ s_0 & 0 \le s_0 \le 1 \\ 1 & s_0 > 1 \end{cases}$$

Proof: set $d\sigma_V^2/ds = 0$. The second derivative is > 0, so this is a global minimum.

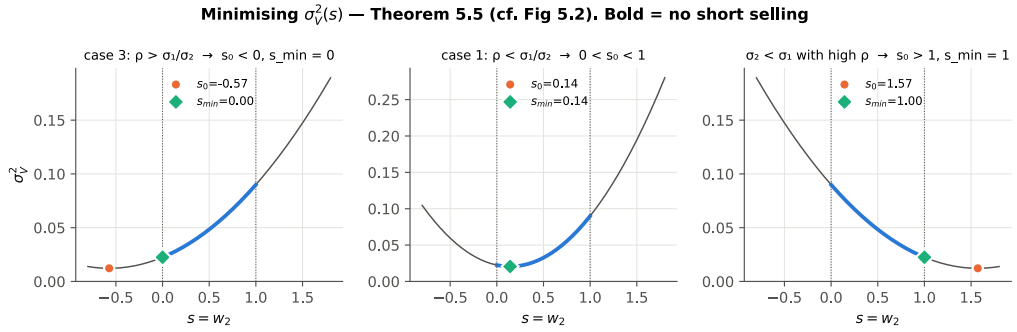


Figure A. *Variance as a function of s, three cases (cf. book Fig. 5.2). Bold segment $= 0 \le s \le 1$. Orange dot = unconstrained minimum s_0; green diamond = minimum without short selling.*

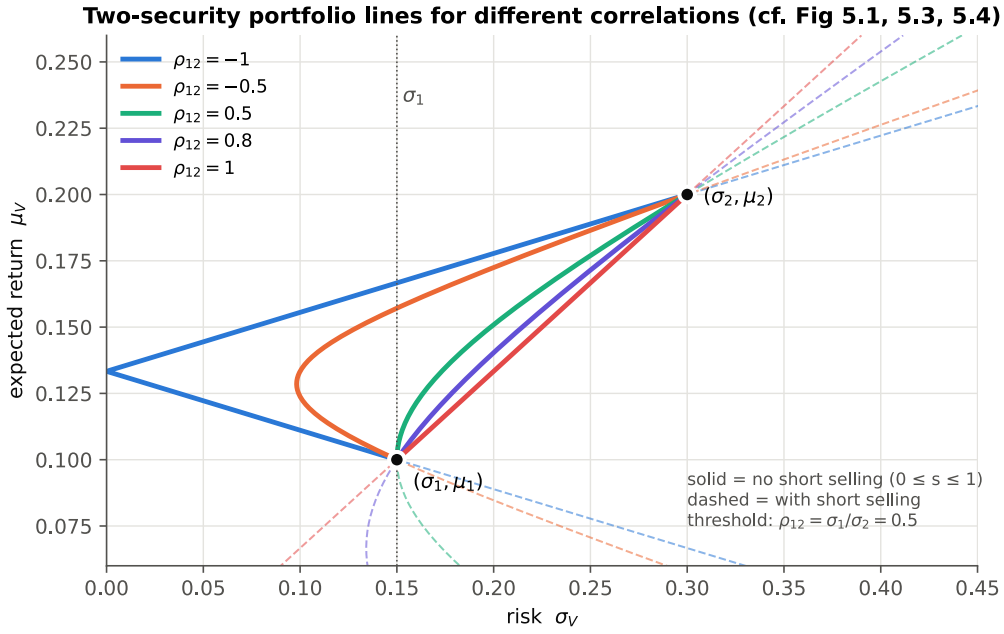


Figure B. *Two-security portfolio lines for several correlations (cf. Figs. 5.1, 5.3, 5.4). Illustrative data: $\mu_1 = 10\%, \sigma_1 = 15\%, \mu_2 = 20\%, \sigma_2 = 30\%$. The lower the correlation, the further the curve bulges to the left (less risk). Here the threshold is $\sigma_1/\sigma_2 = 0.5$.*

Corollary 5.6: when can you beat the less risky asset? (assume $\sigma_1 \le \sigma_2$)

Condition	Result
$-1 \le \rho_{12} < \dfrac{\sigma_1}{\sigma_2}$	there is a portfolio **without** short selling with $\sigma_V < \sigma_1$
$\rho_{12} = \dfrac{\sigma_1}{\sigma_2}$	$\sigma_V \ge \sigma_1$ for **every** portfolio ($s_0 = 0$)
$\dfrac{\sigma_1}{\sigma_2} < \rho_{12} \le 1$	$\sigma_V < \sigma_1$ is possible **only with** short selling (short security 1)

Common mistake. Students often assume "diversification always lowers risk below the safer asset." It does so only when $\rho_{12} < \sigma_1/\sigma_2$. In Figure B, the $\rho = 0.8$ (violet) solid segment never goes left of σ_1.

3.7 One risky + one risk-free security

Proposition 5.7

If security 2 is risk-free ($\sigma_2 = 0$, return r_F):

$$\sigma_V = |w_1|\,\sigma_1, \qquad \mu_V = r_F + w_1(\mu_1 - r_F)$$

On the (σ, μ) plane this is **two straight half-lines** starting at $(0, r_F)$.

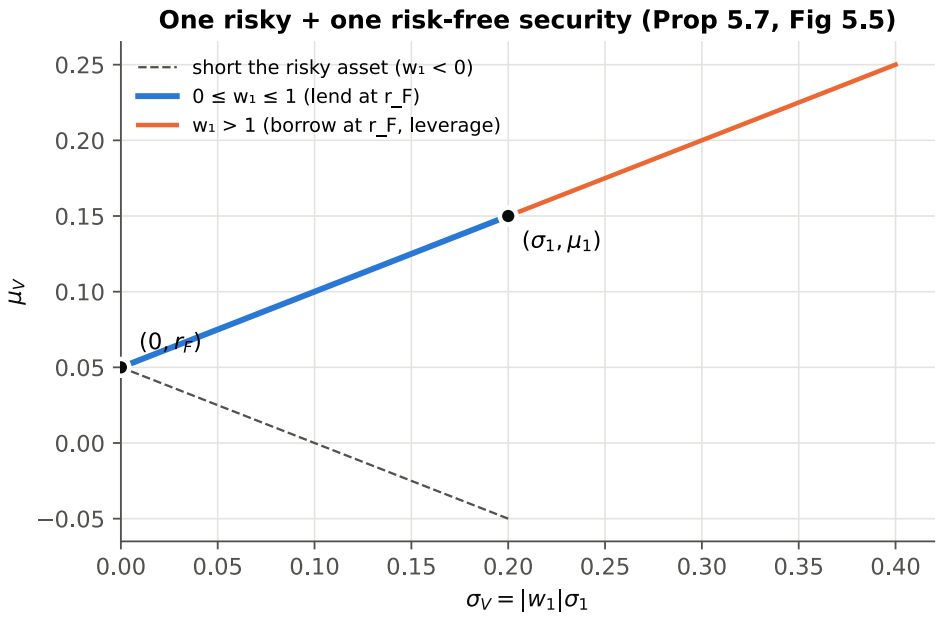


Figure C. *One risky plus one risk-free security (cf. Fig. 5.5). Illustrative:* $r_F = 5\%$, $\mu_1 = 15\%$, $\sigma_1 = 20\%$.

In practice. Splitting money between a liquid/T-bill fund (≈ risk-free) and an equity index fund. Every split lies on the blue segment. Borrowing to buy more equity (margin) takes you onto the orange part.

4 Section 5.3: Several securities

4.1 Matrix set-up

$$w_i = \frac{x_i S_i(0)}{V(0)}, \quad \boldsymbol{w} = [w_1 \cdots w_n], \quad \boldsymbol{m} = [\mu_1 \cdots \mu_n], \quad \boldsymbol{u} = [1 \cdots 1], \quad \boldsymbol{C} = [c_{ij}]$$

Budget constraint

$$\boldsymbol{u}\boldsymbol{w}^T = 1 \tag{5.14}$$

Portfolios that satisfy (5.14) form the **attainable set**.

Proposition 5.8: portfolio mean and variance in matrix form

$$\mu_V = \boldsymbol{m}\boldsymbol{w}^T \tag{5.15}$$

$$\sigma_V^2 = \boldsymbol{w}\boldsymbol{C}\boldsymbol{w}^T = \sum_{i,j=1}^{n} w_i w_j c_{ij} \tag{5.16}$$

With $n = 2$ these reduce exactly to (5.7) and (5.8). Build $\boldsymbol{C}$ from $c_{ij} = \rho_{ij}\sigma_i\sigma_j$.

4.2 The two optimisation problems

1. **Minimum variance portfolio (MVP):** the smallest σ_V^2 over the whole attainable set.
2. **Minimum variance line:** for each target μ_V, the smallest σ_V^2 among portfolios with that expected return.

Both are solved with **Lagrange multipliers**.

Proposition 5.9: Minimum Variance Portfolio

$$\boldsymbol{w}_{MVP} = \frac{\boldsymbol{u}\boldsymbol{C}^{-1}}{\boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{u}^T}$$

Proof sketch: $F = \boldsymbol{w}\boldsymbol{C}\boldsymbol{w}^T - \lambda\boldsymbol{u}\boldsymbol{w}^T$. Setting $\partial F/\partial w_i = 0$ gives $2\boldsymbol{w}\boldsymbol{C} = \lambda\boldsymbol{u}$, so $\boldsymbol{w} = \frac{\lambda}{2}\boldsymbol{u}\boldsymbol{C}^{-1}$. Choose λ so that $\boldsymbol{u}\boldsymbol{w}^T = 1$.

Proposition 5.10: Minimum Variance Line

$$\boldsymbol{w} = \frac{\begin{vmatrix} 1 & \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{m}^T \\ \mu_V & \boldsymbol{m}\boldsymbol{C}^{-1}\boldsymbol{m}^T \end{vmatrix} \boldsymbol{u}\boldsymbol{C}^{-1} + \begin{vmatrix} \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{u}^T & 1 \\ \boldsymbol{m}\boldsymbol{C}^{-1}\boldsymbol{u}^T & \mu_V \end{vmatrix} \boldsymbol{m}\boldsymbol{C}^{-1}}{\begin{vmatrix} \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{u}^T & \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{m}^T \\ \boldsymbol{m}\boldsymbol{C}^{-1}\boldsymbol{u}^T & \boldsymbol{m}\boldsymbol{C}^{-1}\boldsymbol{m}^T \end{vmatrix}}$$

The weights are **linear in** μ_V: $\boldsymbol{w} = \boldsymbol{a}\,\mu_V + \boldsymbol{b}$.

Easier version to memorise. Let

$$A = \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{u}^T, \quad B = \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{m}^T = \boldsymbol{m}\boldsymbol{C}^{-1}\boldsymbol{u}^T, \quad D = \boldsymbol{m}\boldsymbol{C}^{-1}\boldsymbol{m}^T, \quad \Delta = AD - B^2.$$

Then

$$\boxed{\boldsymbol{w} = \frac{(D - B\mu_V)\,\boldsymbol{u}\boldsymbol{C}^{-1} + (A\mu_V - B)\,\boldsymbol{m}\boldsymbol{C}^{-1}}{\Delta}}, \qquad \boxed{\sigma_V^2 = \frac{A\mu_V^2 - 2B\mu_V + D}{\Delta}}$$

The second formula shows the minimum variance line is a **hyperbola** on the (σ, μ) plane. Its vertex is the MVP, at $\mu_{MVP} = B/A$ and $\sigma^2_{MVP} = 1/A$. *(This is the same result as Prop. 5.10, written with letters for the scalars. It is not printed this way in the book.)*

Proof sketch: $G = \boldsymbol{w}\boldsymbol{C}\boldsymbol{w}^T - \lambda\boldsymbol{u}\boldsymbol{w}^T - \mu\boldsymbol{m}\boldsymbol{w}^T$ gives $\boldsymbol{w} = \frac{\lambda}{2}\boldsymbol{u}\boldsymbol{C}^{-1} + \frac{\mu}{2}\boldsymbol{m}\boldsymbol{C}^{-1}$. Substituting into both constraints gives a 2×2 linear system in λ, μ; solve it with Cramer's rule.

Example 5.10 (three securities): the main worked example

i	μ_i	σ_i	correlations
1	0.10	0.28	$\rho_{12} = -0.10$
2	0.15	0.24	$\rho_{23} = 0.20$
3	0.20	0.25	$\rho_{31} = 0.25$

$$\boldsymbol{C} \approx \begin{bmatrix} 0.0784 & -0.0067 & 0.0175 \\ -0.0067 & 0.0576 & 0.0120 \\ 0.0175 & 0.0120 & 0.0625 \end{bmatrix}, \quad \boldsymbol{C}^{-1} \approx \begin{bmatrix} 13.954 & 2.544 & -4.396 \\ 2.544 & 18.548 & -4.274 \\ -4.396 & -4.274 & 18.051 \end{bmatrix}$$

- $\boldsymbol{u}\boldsymbol{C}^{-1} \approx [12.102 \;\; 16.818 \;\; 9.382]$, $A \approx 38.302$
- $\boldsymbol{m}\boldsymbol{C}^{-1} \approx [0.898 \;\; 2.182 \;\; 2.530]$, $D \approx 0.923$, $B \approx 5.609$
- **MVP:** $\boldsymbol{w} \approx [0.316 \;\; 0.439 \;\; 0.245]$, $\mu_V \approx 0.146$, $\sigma_V \approx 0.162$
- **Min-variance line:** $\boldsymbol{w} \approx [1.578 - 8.614\mu_V, \;\; 0.845 - 2.769\mu_V, \;\; -1.422 + 11.384\mu_V]$
- $\sigma_V \approx \sqrt{0.237 - 2.885\mu_V + 9.850\mu_V^2}$

(All of these numbers were re-run in Python and match the book.)

4.3 Visualising the attainable set (Examples 5.11, 5.12)

Picture 1: the weight plane (w_2, w_3; $w_1 = 1 - w_2 - w_3$).

- The vertices of the triangle are single-security portfolios. The edges are two-security portfolios.
- **Inside the triangle** = no short selling. Outside = at least one security is shorted.
- The min-variance line is a **straight line** here, because the weights are linear in μ_V.

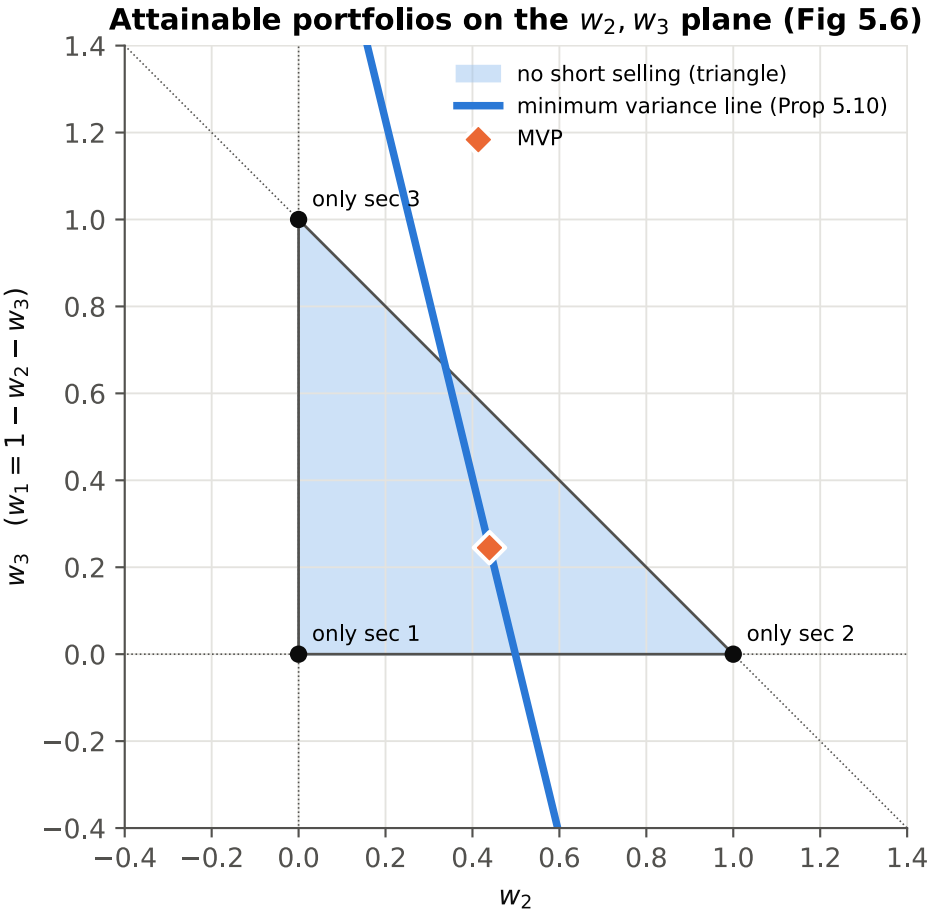


Figure D. *Weight plane (cf. Fig. 5.6), Example 5.10 data.*

Picture 2: the risk–expected return plane, i.e. the Markowitz bullet.

- The whole attainable set (with shorting) is the region to the **right** of the min-variance hyperbola: the **Markowitz bullet**.
- The weight plane gets *folded* along the min-variance line onto this region. So each interior point of the bullet corresponds to **two** portfolios, and each boundary point to exactly one.
- **Without short selling** (Example 5.12) the region shrinks to the blue area. Its boundary is

made of pieces of the two-security curves.

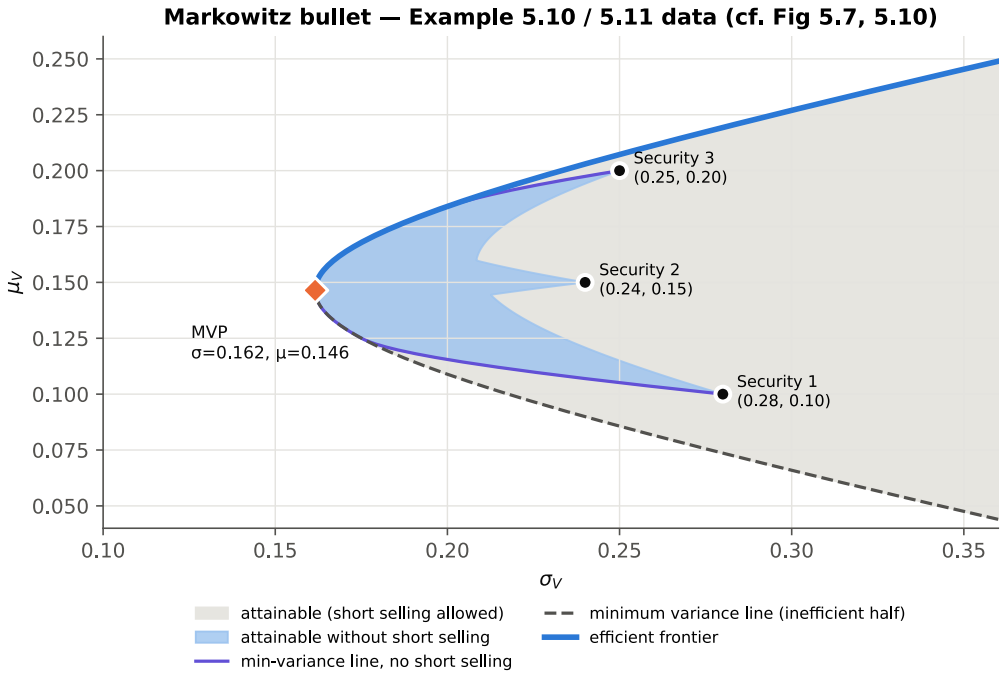


Figure E. *Markowitz bullet (cf. Figs. 5.7, 5.8, 5.10). Grey = attainable with short selling; blue = without. The no-short-selling regions are computed exactly, not sketched.*

4.4 Efficient frontier (§5.3.2)

Definition 5.1: dominance

Security/portfolio 1 **dominates** 2 if

$$\mu_1 \geq \mu_2 \quad \text{and} \quad \sigma_1 \leq \sigma_2$$

Definition 5.2: efficient portfolio / efficient frontier

A portfolio is **efficient** if no other portfolio dominates it. The set of efficient portfolios is the **efficient frontier**.

Key facts:

- The efficient frontier is a **subset of the min-variance line**: the **upper half**, starting at the MVP ($\mu_V \geq \mu_{MVP}$). This is the blue curve in Figure E.
- Remark 5.4: a **dominated** security is not useless. Combined with the dominating one, it can **reduce risk** below either of them (Fig. 5.9).
- Which efficient portfolio to pick depends on the investor's **risk appetite**. Theory says only: *pick something on the frontier.*

Proposition 5.11: two-fund theorem

If $\boldsymbol{w}'$ and $\boldsymbol{w}''$ are any two different portfolios on the min-variance line, then the whole line is

$$\{\, c\boldsymbol{w}' + (1-c)\boldsymbol{w}'' \;:\; c \in \mathbb{R} \,\}$$

In practice. However many securities there are, the min-variance line has the **same shape as a two-security line**. Any two frontier "funds" can replicate every other frontier portfolio.

Proposition 5.12: tangency condition

Every efficient portfolio (other than the MVP) satisfies

$$\gamma\, \boldsymbol{w}\boldsymbol{C} = \boldsymbol{m} - \mu\, \boldsymbol{u} \tag{5.17}$$

for some $\gamma > 0$ and real μ.

Interpretation (Remark 5.5): μ = **intercept** on the μ-axis of the tangent line to the frontier at that portfolio; $\gamma\sigma_V$ = its **slope**.

Proof idea: the tangent line through $(0, \mu)$ has the **maximal slope** $\dfrac{\boldsymbol{m}\boldsymbol{w}^T - \mu}{\sqrt{\boldsymbol{w}\boldsymbol{C}\boldsymbol{w}^T}}$ among lines from $(0, \mu)$ that touch the attainable set. Maximise this subject to $\boldsymbol{u}\boldsymbol{w}^T = 1$ and you get (5.17) with $\gamma = (\mu_V - \mu)/\sigma_V^2$.

This slope is a **Sharpe ratio** (with μ playing the role of r_F). So Prop. 5.12 is the maths behind "the tangency portfolio maximises the Sharpe ratio."

5 Section 5.4: Capital Asset Pricing Model (CAPM)

Motivation: with $n = 1000$ securities, $\boldsymbol{C}$ has 10^6 entries. Estimating and inverting it is hard. CAPM gives a simpler structure.

Assumption: all investors use the **same** μ_i, σ_i, ρ_{ij}, so everyone computes the **same efficient frontier**. They differ only in risk appetite.

5.1 Capital Market Line (CML)

- Add a risk-free asset with return r_F. Combining r_F with any risky portfolio gives straight half-lines from $(0, r_F)$ (Prop. 5.7).
- The new efficient frontier is the half-line from $(0, r_F)$ that is **tangent to the Markowitz bullet**. This is the **Capital Market Line**.
- The tangency point (σ_M, μ_M) is the **market portfolio** M. Every investor holds risky assets in the **same proportions** as M, so M must contain every risky security in proportion to its share of total market value.

- The argument needs r_F to be not too high (below μ_{MVP}), so that a tangent exists.

Capital Market Line

$$\mu = r_F + \frac{\mu_M - r_F}{\sigma_M}\,\sigma \tag{5.18}$$

Risk premium (on the CML) $= \dfrac{\mu_M - r_F}{\sigma_M}\,\sigma$. The slope $\dfrac{\mu_M - r_F}{\sigma_M}$ is the **market price of risk** (the market's Sharpe ratio).

Computing the market portfolio (Example 5.13)

From (5.17) with $\mu = r_F$:

$$\gamma \boldsymbol{w}_M = (\boldsymbol{m} - r_F \boldsymbol{u})\boldsymbol{C}^{-1}, \qquad \text{then normalise so that } \boldsymbol{u}\boldsymbol{w}_M^T = 1$$

Example 5.13 (Example 5.10 data, $r_F = 5\%$): $\gamma \boldsymbol{w} \approx [0.293 \;\; 1.341 \;\; 2.061]$, so $\gamma \approx 3.694$ and

$$\boldsymbol{w}_M \approx [0.079 \;\; 0.363 \;\; 0.558]$$

(The book doesn't print these, but the code gives $\mu_M \approx 0.174$ and $\sigma_M \approx 0.183$.)

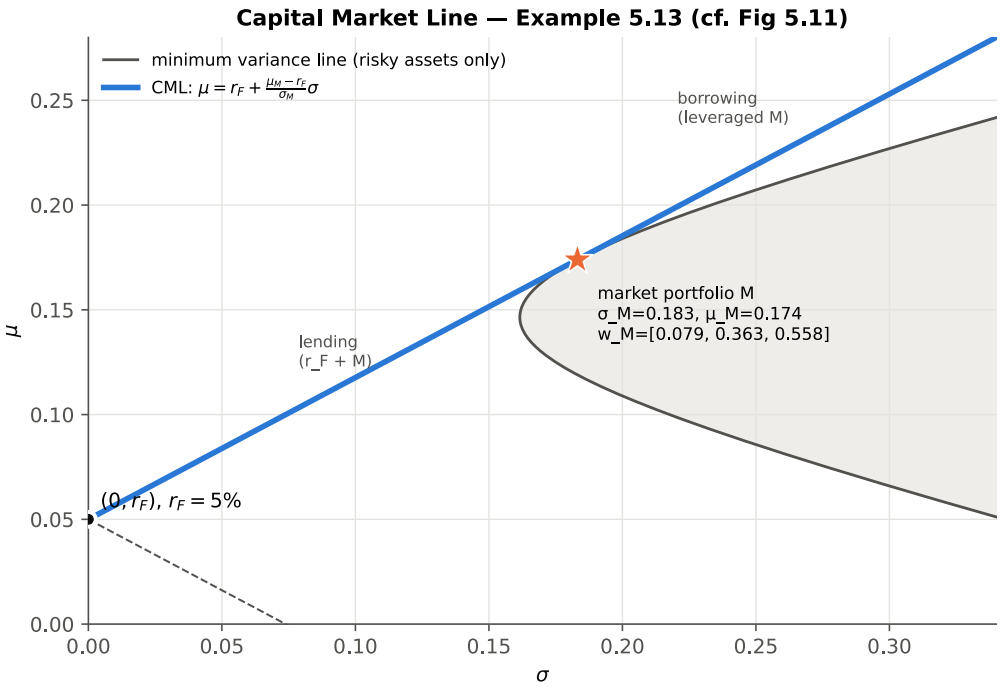


Figure F. *Capital Market Line (cf. Fig. 5.11).*

In practice. This is the theory behind **index investing**. The market portfolio is approximated by a broad index (e.g. Nifty 50 / Nifty 500 in India, S&P 500 in the US). A conservative investor holds some index fund plus liquid/G-sec funds (lending part of the CML). An aggressive one leverages the index (borrowing part).

5.2 Beta factor

Plot K_V against K_M across scenarios and fit the **line of best fit** (regression / **characteristic line**):

$$y = \beta_V x + \alpha_V$$

Residual: $\varepsilon_V = K_V - (\alpha_V + \beta_V K_M)$. Minimise $E(\varepsilon^2)$ over α, β. That gives the normal equations

$$\alpha_V E(K_M) + \beta_V E(K_M^2) = E(K_V K_M), \qquad \alpha_V + \beta_V E(K_M) = E(K_V).$$

Beta and alpha (Definition 5.3)

$$\beta_V = \frac{\mathrm{Cov}(K_V, K_M)}{\sigma_M^2}, \qquad \alpha_V = \mu_V - \beta_V \mu_M$$

Risk decomposition

$$\sigma_V^2 = \underbrace{\mathrm{Var}(\varepsilon_V)}_{\text{residual / diversifiable}} + \underbrace{\beta_V^2 \sigma_M^2}_{\text{systematic / undiversifiable}}$$

- **Diversifiable (residual) risk** can be removed by diversifying. For the market itself, $\mathrm{Var}(\varepsilon_M) = 0$.
- **Systematic risk** ($\propto \beta_V^2$) **cannot** be diversified away. In CAPM, **only systematic risk earns a premium.**
- $\beta > 0 \rightarrow$ tends to move with the market. $\beta < 0 \rightarrow$ tends to move against it.
- **Beta is linear in weights**: $\beta_V = w_1\beta_1 + \cdots + w_n\beta_n$. Also $\beta_M = 1$ and $\beta_F = 0$.

In practice. This is why 20–30 well-spread stocks capture most of the diversification benefit, and why a stock's **beta** (not its total volatility) drives its cost of equity in a DCF / WACC.

5.3 Security Market Line (SML)

Start from $\gamma \boldsymbol{w}_M \boldsymbol{C} = \boldsymbol{m} - \mu \boldsymbol{u}$. For any portfolio $\boldsymbol{w}_V$:

$$\beta_V = \frac{\boldsymbol{w}_M \boldsymbol{C} \boldsymbol{w}_V^T}{\boldsymbol{w}_M \boldsymbol{C} \boldsymbol{w}_M^T} = \frac{\mu_V - \mu}{\mu_M - \mu}.$$

Plugging in the risk-free asset ($\beta_F = 0$, return r_F) shows $\mu = r_F$.

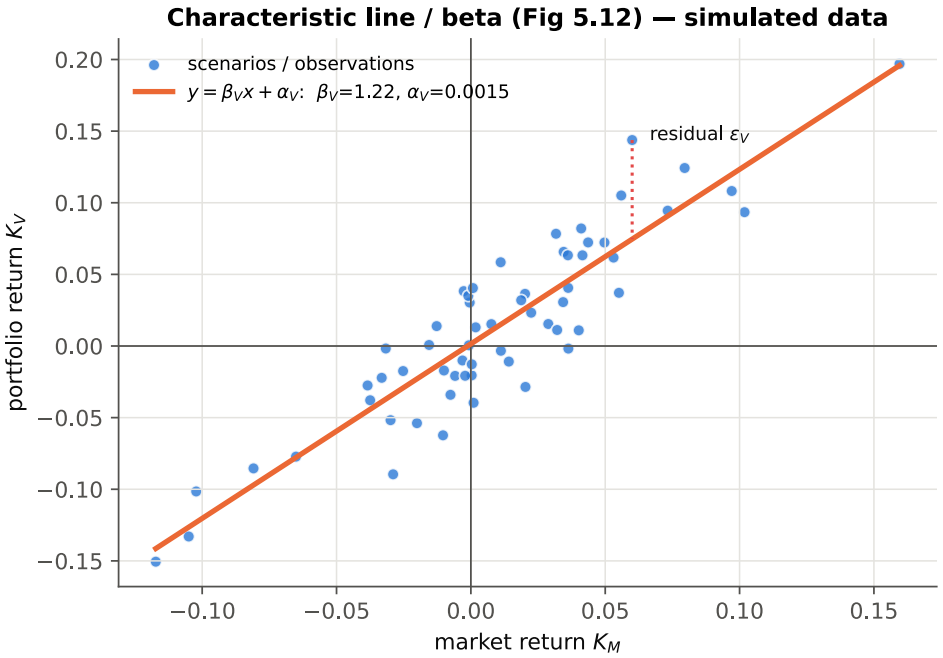


Figure G. *Characteristic line (cf. Fig. 5.12). Simulated monthly returns with true $\beta = 1.3$. The fitted β differs because of noise, which is the estimation problem you meet in practice.*

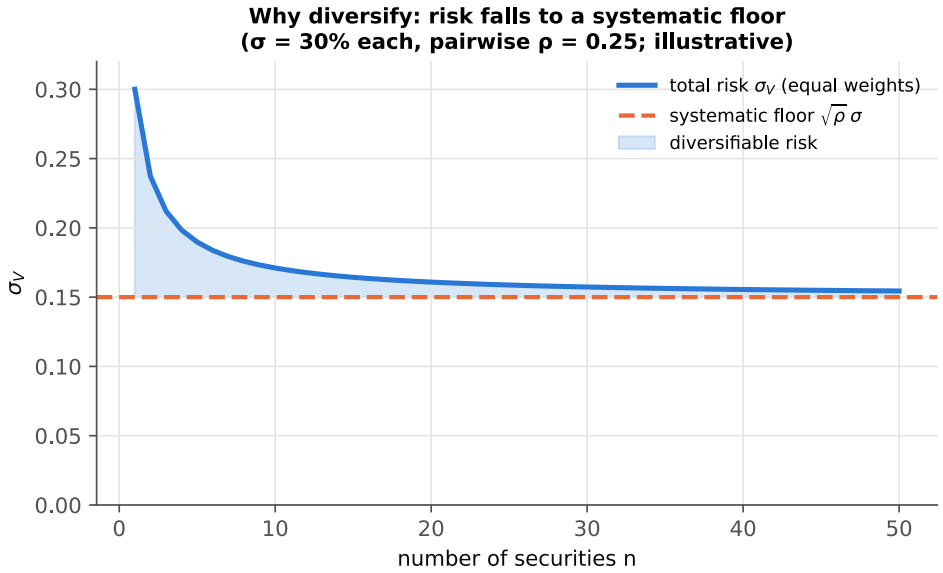


Figure H. *Diversification and the systematic floor (illustrative,* beyond the book*): an equal-weighted portfolio of n securities, each with $\sigma = 30\%$ and pairwise $\rho = 0.25$. Then $\sigma_V^2 = \sigma^2/n + (1 - 1/n)\rho\sigma^2$. The diversifiable part goes to zero as n grows; the systematic floor $\sqrt{\rho}\,\sigma$ stays.*

Theorem 5.13: Security Market Line (the CAPM equation)

$$\mu_V = r_F + (\mu_M - r_F)\,\beta_V \tag{5.19}$$

Risk premium $= (\mu_M - r_F)\beta_V$: compensation for **systematic** risk only.

	CML (5.18)	**SML (5.19)**
Axes	(σ, μ)	(β, μ)
Risk measure	total risk σ	systematic risk β
Applies to	**efficient** portfolios only	**every** security and portfolio
Slope	$(\mu_M - r_F)/\sigma_M$	$\mu_M - r_F$
Passes through	$(0, r_F)$ and (σ_M, μ_M)	$(0, r_F)$ and $(1, \mu_M)$

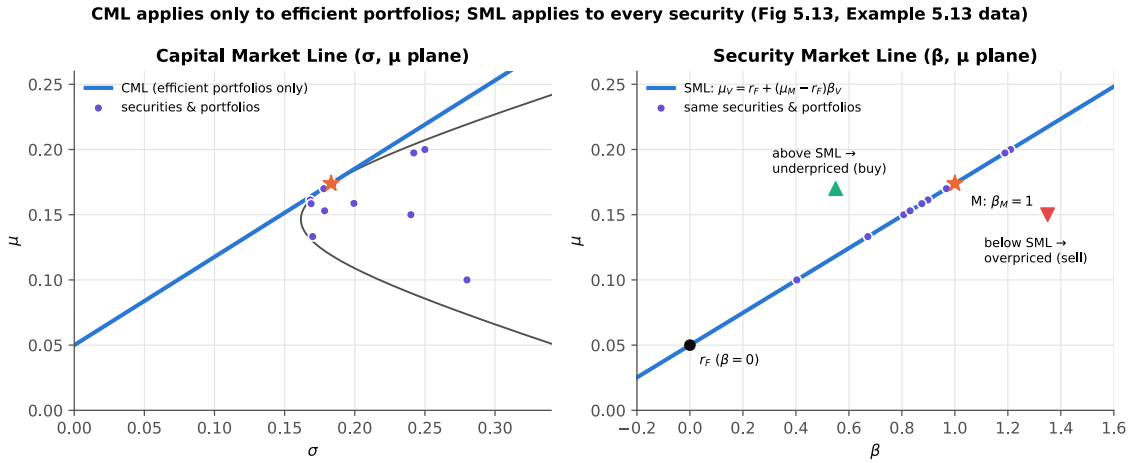


Figure I. *CML vs SML (cf. Fig. 5.13). The same securities and portfolios: scattered **inside** the bullet on the left, all **on** the SML on the right.*

5.4 Equilibrium and mispricing

- In equilibrium everyone holds M (plus borrowing or lending), so supply equals demand.
- If new information gives
  - $\mu_V > r_F + (\mu_M - r_F)\beta_V$ (**above SML**): the security is **underpriced**. Investors buy, the price rises, and μ_V falls back to the line.
  - $\mu_V < r_F + (\mu_M - r_F)\beta_V$ (**below SML**): the security is **overpriced**. Investors sell, the price falls, and μ_V rises.

In practice. The gap $\mu_V - [r_F + (\mu_M - r_F)\beta_V]$ is **Jensen's alpha**, used to judge whether a fund manager added value. (That is related to, but not the same as, the regression intercept α_V

above.)

Common point of characteristic lines: under CAPM, $\alpha_V = \mu_V - \beta_V \mu_M = r_F(1 - \beta_V)$. So every characteristic line $y = r_F + \beta_V(x - r_F)$ passes through the **common point** (r_F, r_F).

6 One-page formula sheet

#	Formula	Where
1	$\mathrm{Var}(aK) = a^2\mathrm{Var}(K)$, $\sigma_{aK} = \lvert a\rvert\sigma_K$	§5.1
2	$w_i = x_i S_i(0)/V(0)$, $\sum w_i = 1$	(5.1), (5.14)
3	$K_V = \sum w_i K_i$	(5.2)
4	$e^{k_V} = \sum w_i e^{k_i}$	(5.3)
5	$\mu_V = w_1\mu_1 + w_2\mu_2$	(5.7)
6	$\sigma_V^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$	(5.8)
7	$\rho = 1$: $w_1 = -\frac{\sigma_2}{\sigma_1 - \sigma_2}$; $\rho = -1$: $w_1 = \frac{\sigma_2}{\sigma_1 + \sigma_2}$ (zero risk)	(5.9), (5.10)
8	$s_0 = \dfrac{\sigma_1^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$, $s = w_2$	(5.13)
9	Risk-free mix: $\sigma_V = \lvert w_1\rvert\sigma_1$	Prop. 5.7
10	$\mu_V = \boldsymbol{m}\boldsymbol{w}^T$, $\sigma_V^2 = \boldsymbol{w}\boldsymbol{C}\boldsymbol{w}^T$	(5.15), (5.16)
11	MVP: $\boldsymbol{w} = \dfrac{\boldsymbol{u}\boldsymbol{C}^{-1}}{\boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{u}^T}$	Prop. 5.9
12	Min-var line: $\boldsymbol{w} = \dfrac{(D - B\mu_V)\boldsymbol{u}\boldsymbol{C}^{-1} + (A\mu_V - B)\boldsymbol{m}\boldsymbol{C}^{-1}}{AD - B^2}$	Prop. 5.10
13	$\sigma_V^2 = \dfrac{A\mu_V^2 - 2B\mu_V + D}{AD - B^2}$	from Prop. 5.10
14	Efficient portfolio: $\gamma\boldsymbol{w}\boldsymbol{C} = \boldsymbol{m} - \mu\boldsymbol{u}$, $\gamma > 0$	(5.17)
15	Market portfolio: $\boldsymbol{w}_M \propto (\boldsymbol{m} - r_F\boldsymbol{u})\boldsymbol{C}^{-1}$, normalised	§5.4.1
16	CML: $\mu = r_F + \dfrac{\mu_M - r_F}{\sigma_M}\sigma$	(5.18)
17	$\beta_V = \dfrac{\mathrm{Cov}(K_V, K_M)}{\sigma_M^2}$, $\alpha_V = \mu_V - \beta_V\mu_M$	Def. 5.3
18	$\sigma_V^2 = \mathrm{Var}(\varepsilon_V) + \beta_V^2\sigma_M^2$	§5.4.2
19	$\beta_V = \sum w_i\beta_i$	§5.4.2
20	SML: $\mu_V = r_F + (\mu_M - r_F)\beta_V$	(5.19)

where $A = \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{u}^T$, $B = \boldsymbol{u}\boldsymbol{C}^{-1}\boldsymbol{m}^T$, $D = \boldsymbol{m}\boldsymbol{C}^{-1}\boldsymbol{m}^T$.

7 Common mistakes and quick checks

1. **Weights must sum to 1**, and they can be negative when shorting is allowed. Always check $\boldsymbol{u}\boldsymbol{w}^T = 1$ in your final answer.

2. **Covariance from correlation:** $c_{ij} = \rho_{ij}\sigma_i\sigma_j$. Don't put ρ directly into $\boldsymbol{C}$.

3. If the problem gives **variances**, take square roots before using $\rho\sigma_1\sigma_2$.

4. **"No short selling"** $\rightarrow$ clip s_0 to $[0, 1]$ (Theorem 5.5). With $n \geq 3$, the long-only frontier is **not** given by Prop. 5.10; it needs constrained optimisation.

5. **The efficient frontier is only the upper half** of the min-variance line.

6. **CML vs SML:** use σ on the CML (efficient portfolios only) and β on the SML (anything).

7. Sanity checks: $\beta_M = 1$, $\beta_F = 0$, $\sigma^2_{MVP} = 1/A$, $\mu_{MVP} = B/A$.

8. When you apply Corollary 5.6, label the securities so that $\sigma_1 \leq \sigma_2$ first.